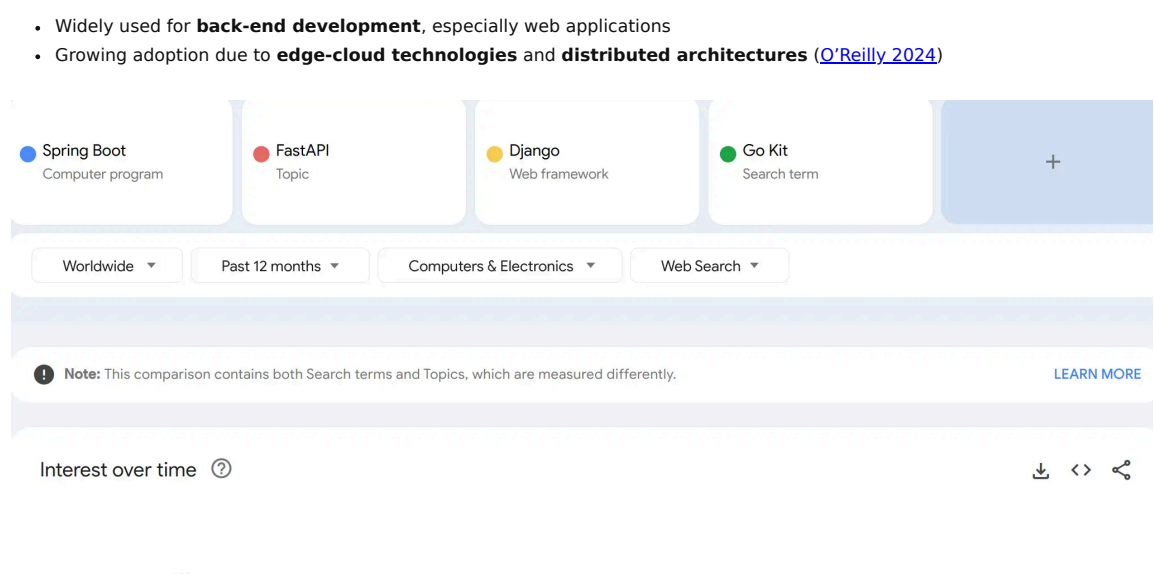
The Spring Ecosystem

Why Spring is Popular

- Widely used for **back-end development**, especially web applications
- Growing adoption due to **edge-cloud technologies** and **distributed architectures** (O'Reilly 2024)



Note: This comparison contains both Search terms and Topics, which are measured differently.

Spring Boot

- Rapid application setup with **minimal configuration**

Key Features:

- Embedded web servers → no WAR deployment needed
- Starter dependencies → simplified builds
- Auto-configuration → reduces boilerplate
- Production-ready → metrics, health checks, externalized config
- Minimal XML configuration

Spring Initializr

- Quick project generation with **pre-configured Spring Boot setup**

Key Features:

- Dependency selection → choose starters like Web, JPA, Security
- Project metadata → group, artifact, packaging, Java version
- Multiple build tools → Maven or Gradle support
- Customizable structure → ready-to-run project templates
- Seamless integration → works with IDEs like IntelliJ and Eclipse

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Spring Core

- Provides **essential building blocks** for Spring applications

Key Features:

- Core: Dependency Injection, Events, Resources, i18n, Validation, Data Binding, Type Conversion, SpEL, AOP
- Testing: Mock objects, TestContext, Spring MVC Test, WebTestClient
- Data Access: Transactions, DAO support, JDBC, ORM, XML Marshalling
- Integration: JMS, JMX, Email, Scheduling, Caching, Observability

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Spring Data

- Simplifies **data access across multiple data stores**

Key Features:

- Repository abstraction & custom object mapping
- Dynamic query derivation from method names
- Domain base classes with common properties
- Transparent auditing (created/last modified)

Spring Cloud

- Supports **cloud-native and distributed architectures**

Key Features:

- Distributed configuration
- Service discovery & routing
- Load balancing
- Resiliency & fault tolerance tools

Spring Security

- Provides **authentication, authorization, and protection mechanisms**

Key Features:

- Authentication & authorization
- Password encoding & user management
- Session & security context management
- CSRF & CORS protection
- HTTP headers hardening
- OAuth2 & OpenID Connect integration
- Method-level security & JWT stateless APIs

Resources

- Spring Framework